

AP Precalculus Unit 2 Wkst C
Graphing Tangent and Cotangent

Name: _____

- a) convert each function to AP (phase shift) form and state the phase shift**
b) graph one period of each function
c) is the function increasing or decreasing over the period?
d) using interval notation, describe the rate of change from the left asymptote to the right asymptote, including the points at which it changes, if any

1. $y = 2 \tan (2\pi x - \pi) + 1$

6. $y = \cot (\pi x - \frac{\pi}{2}) - 3$

2. $y = -3 \tan (\frac{\pi}{2}x + \frac{\pi}{2}) - 2$

7. $y = -5 \cot (\frac{\pi}{4}x + \frac{\pi}{2}) + 2$

3. $y = \tan (2x - \frac{\pi}{6}) + 5$

8. $y = 3 \cot (\frac{\pi}{2}x + \frac{\pi}{2}) - 2$

4. $y = 7 \tan (x + \frac{\pi}{12})$

9. $y = -\cot (2x - \frac{\pi}{4}) + 4$

5. $y = -2 \tan (4\pi x + 3\pi) - 4$

10. $y = -\frac{1}{2} \cot (3\pi x - \pi) - 2$